

AWMA-MoE: Attention-Guided Watermark Adapter with MoE for Latent Diffusion Models

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Abstract

With the evolving generative models, generated images are closer to reality, raising concerns about information authenticity and malicious misuse. Invisible watermarks offer a practical approach to detecting and tracing them. However, while image watermarking inevitably introduces quality degradation, most existing methods primarily focus on improving watermark robustness. To address this limitation, we propose AWMA-MoE, a framework that enhances the quality of generated images while preserving strong watermark robustness. Specifically, we design an attention-based adapter that adaptively embeds watermarks with spatially varying strengths across image regions. Building upon this, we introduce an MoE architecture that leverages diverse experts to further improve image quality while retaining watermark robustness. Experiments demonstrate that AWMA-MoE can reduce the distortion of generated images and exhibit competitive watermark performance, thus striking an improved balance for watermarking generated image tasks and better linking post-hoc and in-generation methods.

CCS Concepts

• Security and privacy → Human and societal aspects of security and privacy.

Keywords

Latent Diffusion Model; Image Generation; Watermarking

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1 Introduction

With the rapid advancement and widespread accessibility of generative models, a substantial amount of AI-generated content (AIGC) has emerged. Among these, text-to-image diffusion models are capable of generating high-resolution images, which may raise several concerns. Generative models may produce similar outputs for different prompts, leading to user copyright disputes. In addition, malicious actors might create and disseminate realistic fake images, posing a threat to public safety. Digital watermarking offers a solution to these problems.

For AIGC images, existing watermarking methods primarily consist of post-hoc and in-generation methods. Post-hoc methods add watermarks to the images after generation, while in-generation ones integrate watermarking into the generation pipeline — making them more suitable for generating scenes. Stable Signature introduced a paradigm that fine-tunes the decoder of the Latent Diffusion Model (LDM) using a pre-trained watermark decoder, rapidly incorporating the watermark into the model.

Subsequently, a range of more comprehensive watermarking methods has emerged, many of which opt to train a new watermark decoder to extract watermarks [8]. However, there are still unique advantages to using pre-trained watermark decoders. It enables the generation system to share one watermark decoder with traditional watermarking systems, eliminating the need to develop decoders for new frameworks. It should also be noted that watermarking will degrade image quality, introducing artifacts or altering the distribution. However, previous methods predominantly emphasized robustness, downplaying the significance of image quality.

In this paper, we propose the Attention-Guided Watermark Adapter with Mixture of Experts (AWMA-MoE), a method trained with a pre-trained watermark decoder for embedding watermarks in images generated by LDM. We insert several watermark adapters equipped with *AttentionModules* into the LDM decoder. These adapters can fuse the image and watermark features during generation and are used as experts to construct a Mixture of Experts (MoE) system with a selector. Specifically, *AttentionModule* computes a mask based on image features, which regulates the intensity of watermarks in different regions to impose less impact on image quality. The selector chooses an optimal expert based on image features for watermarking, thereby enhancing both image quality

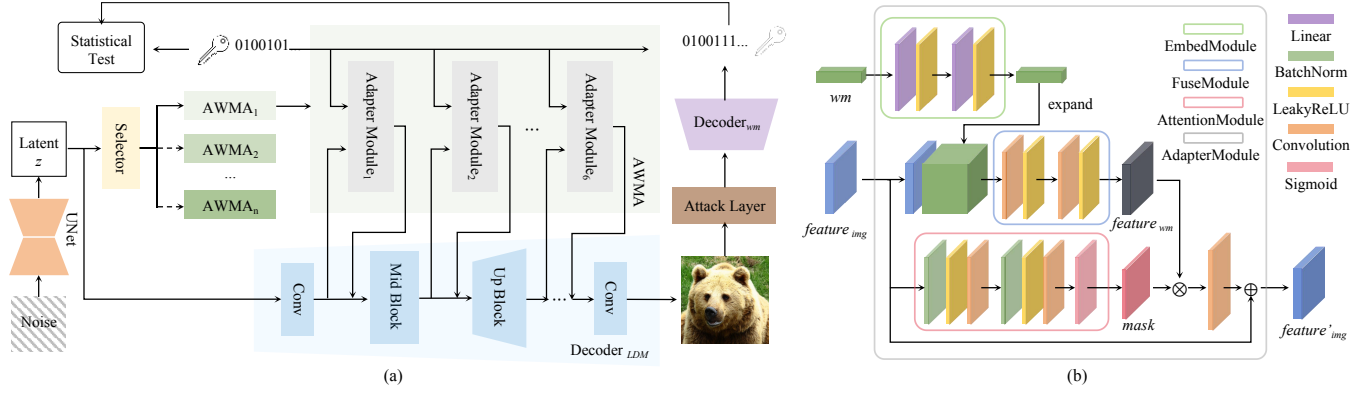


Figure 1: The proposed method. (a) The framework of AWMA-MoE. (b) The structure of the designed watermark adapter.

and maintaining watermark performance. We evaluated AWMA-MoE on two datasets, and the results indicate that it can generate watermarked images of high quality while preserving watermark robustness. The contributions are as follows:

- We design a watermark adapter with *AttentionModules* that compute masks from image features to control the watermarking intensity for different regions.
- We incorporate an MoE framework into the watermark embedding process. The selector chooses suitable experts based on image features for watermarking.
- The results on two datasets show that AWMA-MoE can significantly improve image quality and maintain watermark robustness with the same watermark decoder.

2 Method

2.1 Overview

Figure 1 (a) shows the framework of AWMA-MoE. It is a plug-and-play watermarking model appended to the LDM decoder, embedding the watermark during the image decoding process. AWMA-MoE features a Mixture of Experts structure, which comprises a selector and multiple watermark adapters called AWMA, with each AWMA being an independent expert.

2.2 AWMA

The Latent Diffusion Model (LDM) includes a pair of VAE. The encoder $Encoder_{LDM}$ maps the image x to the latent space z , $z = Encoder_{LDM}(x)$, and the decoder $Decoder_{LDM}$ can restore z to the image x' , $x' = Decoder_{LDM}(z)$. Therefore, for LDM, the denoising process outputs a latent vector z , which is then upsampled by $Decoder_{LDM}$ through six successive blocks.

To best preserve the performance of the generative models, inserting adapters into $Decoder_{LDM}$ for watermarking is a viable approach. Only the adapter parameters are trained, while $Decoder_{LDM}$ remains frozen to produce a high-quality watermarked image $x'_w = Decoder'_{LDM}(z)$, where $Decoder'_{LDM}$ denotes the LDM decoder integrated with the watermark adapter.

We insert one *AdapterModule* before each of the blocks. These six *AdapterModule* _{i} , $i = 1, 2, \dots, 6$ form the watermark adapter $Adapter_{wm}$. *AdapterModule* _{i} takes as inputs the image features

f_i from the preceding block and the watermark information $w \in \{0, 1\}^l$, a binary string of length l . Specifically, the watermark information w is first embedded and fused with the image features f_i to yield the watermark features y_i . Then, y_i is added to f_i via a residual structure, producing the watermarked image features f'_i , which are passed on to the subsequent processing block.

$$\begin{aligned} y_i &= AdapterModule_i(f_i, w), \\ f'_i &= f_i + y_i. \end{aligned} \quad (1)$$

According to ARWGAN [6], injecting watermarks in different regions of images leads to varying degrees of image distortion and watermark robustness. Consequently, we incorporate *AttentionModules* within *AdapterModule*, constituting AWMA. This module computes an attention mask to adjust watermark intensity across image feature regions and improve the quality of generated images.

$$\begin{aligned} y_i &= FuseModule_i(f_i, EmbedModule_i(w)), \\ m_i &= AttentionModule_i(f_i), \\ f'_i &= m_i \odot y_i + f_i. \end{aligned} \quad (2)$$

where m_i is the mask generated based on the image features f_i , and $\odot$ represents element-wise multiplication. Figure 1 (b) illustrates the structure of *AdapterModule*.

2.3 Mixture of Experts

Due to variations in the training settings (e.g., initialization and training data), AWMA shows different levels of image quality and watermark robustness for the same image. Mixture of Experts (MoE) exactly utilizes expert diversity to enhance the model's performance. Therefore, we build an MoE model, AWMA-MoE, which comprises a selector and k AWMA, $AWMA_i$, $i = 1, \dots, k$.

For efficiency in AWMA-MoE, we design a single selector whose results apply to all AWMA. The selector, mainly comprising 5 convolutional blocks, maps the channel dimension of z to the number of experts, aggregates the features through global average pooling, and computes weights via Softmax. As shown in Figure 1 (a), during decoding z , the selector takes z and selects the best $AWMA_p$ to embed the watermark, producing higher-quality images x'_w .

$$\begin{aligned} p &= \operatorname{argmaxSoftmax}((\operatorname{selector}(z))), \\ x'_w &= Decoder_{LDM, AWMA_p}(z). \end{aligned} \quad (3)$$

Table 1: Comparison between previous methods and ours regarding image quality and watermark robustness. Results are obtained on MS-COCO 2017 validation set and CLIC 2020 test set, and the image size is 256×256 . We report the bit accuracy under several attacks and a Combination (Crop 0.5, Jpeg 80, Brightness 1.5). Best results are in bold, second-best are underlined.

Method		Image Quality		Embed. Time	Bit Accuracy $\uparrow$						
		PSNR/SSIM $\uparrow$	LPIPS/SIFID $\downarrow$		None	Crop 0.1	Jpeg 80	Brig. 2.0	Cont. 2.0	Resi. 0.7	Comb.
COCO 2017											
Post-hoc	dctDwtSvd	<u>38.2/0.99</u>	<u>0.013/0.014</u>	27	1.00	0.50	0.91	0.55	0.55	0.51	0.50
	HiDDeN	36.8/0.96	0.024/0.057	5	<u>0.99</u>	0.89	0.88	0.98	<u>0.99</u>	<u>0.90</u>	0.83
	MBRS	42.7/0.98	0.005/0.001	108	1.00	<u>0.74</u>	1.00	0.98	1.00	0.51	0.59
	SSL	35.0/0.94	0.072/0.114	5808	1.00	0.61	<u>0.98</u>	<u>0.88</u>	0.91	0.95	<u>0.75</u>
In-gen	LaWa	31.0/0.89	0.029/0.012	11	1.00	<u>0.74</u>	<u>0.99</u>	0.98	1.00	<u>0.93</u>	0.57
	RoSteALS	<u>32.6/0.97</u>	<u>0.018/0.009</u>	7	1.00	0.49	1.00	0.94	0.96	0.52	0.49
	Stable Signature	29.7/0.89	0.034/0.056	0	1.00	0.93	0.92	<u>0.97</u>	<u>0.98</u>	0.98	<u>0.91</u>
	ours	34.6/0.96	0.009/0.014	16	<u>0.99</u>	0.93	0.93	<u>0.97</u>	<u>0.98</u>	0.98	0.93
CLIC 2020											
Post-hoc	dctDwtSvd	<u>38.6/0.99</u>	<u>0.015/0.017</u>	26	1.00	0.50	0.92	0.54	0.54	0.51	0.50
	HiDDeN	36.9/0.96	0.027/0.055	5	<u>0.99</u>	0.88	0.88	0.97	<u>0.99</u>	0.90	0.82
	MBRS	43.0/0.98	0.006/0.001	105	1.00	<u>0.73</u>	1.00	0.97	1.00	0.50	0.58
	SSL	35.0/0.94	0.082/0.122	5819	1.00	0.61	<u>0.98</u>	<u>0.86</u>	0.91	0.95	<u>0.74</u>
In-gen	LaWa	31.0/0.89	0.028/0.012	11	1.00	<u>0.74</u>	<u>0.99</u>	0.98	1.00	0.92	0.57
	RoSteALS	<u>33.4/0.98</u>	<u>0.017/0.009</u>	8	1.00	0.50	1.00	0.93	0.96	0.52	0.49
	Stable Signature	30.2/0.90	0.033/0.056	0	<u>0.99</u>	0.92	0.91	<u>0.95</u>	<u>0.98</u>	<u>0.97</u>	<u>0.90</u>
	ours	34.8/0.96	0.009/0.015	16	<u>0.99</u>	0.92	0.92	<u>0.95</u>	0.97	0.98	0.92

where p represents the index of the selected AWMA expert, and $Decoder_{LDM,AWMA_p}(z)$ means that z is decoded by $Decoder_{LDM}$ combined with $AWMA_p$.

2.4 Training

The training is divided into two stages. A pre-trained watermark decoder $Decoder_{wm}$ is used to extract watermarks from watermarked images. In AWMA stage, we freeze the parameters of $Decoder_{wm}$ and $Decoder_{LDM}$ and only train the AWMA. In MoE stage, we construct MoE model with a selector and various pre-trained AWMA, and jointly update the selector while fine-tuning the AWMA.

We employ image distortion loss and perceptual loss to preserve visual and perceptual similarity between original and watermarked images. For distortion loss, we adopt the mean absolute error loss (MAE). Regarding perceptual loss, we utilize LPIPS [12] and VGG-Waston [3] losses to minimize perceptual differences. These three losses together form the image quality loss.

$$L_{img} = L_{mae} + L_{lips} + \alpha L_{vgg}. \quad (4)$$

where α is set to 0.1 to balance the larger value of L_{vgg} . For the watermark loss, Binary Cross-Entropy (BCE) loss is used.

$$\begin{aligned} w' &= Decoder_{wm}(x'_w), \\ L_{wm} &= BCE(w, w'). \end{aligned} \quad (5)$$

Combining two parts of the losses, we define the overall loss function as follows:

$$L_{total} = L_{img} + L_{wm}. \quad (6)$$

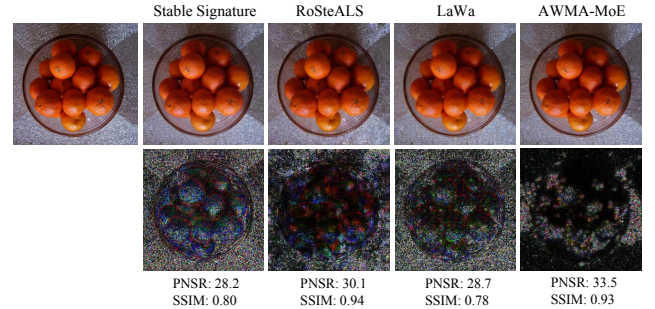


Figure 2: Visual quality comparison with baselines.

In addition, we introduce an attack simulation layer to the framework, including common image attacks, to enhance watermark robustness from the perspective of features. The attack layer is inactive at first and is activated once the bit accuracy reaches 0.9.

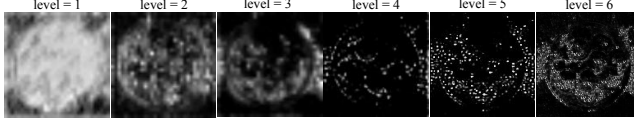
3 Experiments

3.1 Setup

For the generative model, we adopt Stable Diffusion 2.1 [10] along with its kl-f8 VAE. For the pre-trained watermark decoder, we select HiDDeN and the 48-bit checkpoint provided by Stable Signature. For both training stages, we employ the AdamW optimizer and a batch size of 4 on a 24 GB GPU for 256×256 images. The learning rate is set to $1e-4$, and the number of MoE's experts is set to 3. We

Table 2: Ablation study on *AttentionModule* and MoE.

AM	MoE	PSNR / SSIM $\uparrow$	LPIPS / SIFID $\downarrow$	$Acc_{Comb.} \uparrow$
✓	✓	34.6 / 0.96	0.009 / 0.014	0.93
✓	×	33.3 / 0.95	0.013 / 0.028	0.93
×	✓	32.6 / 0.94	0.016 / 0.025	0.94
×	×	31.9 / 0.93	0.020 / 0.037	0.94

**Figure 3: Visualization of multilevel masks.**

train all models on the MS-COCO 2017 [7] training set and derive test results from its validation set and the CLIC 2020 [11] test set.

Concerning image quality, we report PSNR, SSIM, LPIPS, and SIFID between images before and after watermarking. Regarding watermark robustness, we report the average bit accuracy rate. The watermark embedding time is also presented to reflect efficiency.

We compare AWMA-MoE with prior in-generation (Stable Signature [4], RoSteALS [1], LaWa [9]) and post-hoc baselines (frequency-based: dctDwtSvd [2]; encoder-decoder-based: HiDDeN [13], MBRS; optimization-based: SSL [5]). All methods embed 48-bit binary strings into 256×256 generated images.

3.2 Main Results

The quantitative results are shown in Table 1. The left section of Table 1 presents image quality indicators. Among in-generation methods, AWMA-MoE achieves the highest PSNR of 34.6 and the lowest LPIPS of 0.009 on the COCO dataset. Compared to Stable Signature with the same watermark decoder, the PSNR and SSIM of AWMA-MoE increase by 4.9 and 0.07, whereas the LPIPS and SIFID decrease by 0.025 and 0.042. The right section of Table 1 shows the robustness of the methods. Compared to other methods, AWMA-MoE achieves the highest bit accuracy of around 0.93 for cropping, 0.98 for resizing, and about 0.93 for combined attacks. AWMA-MoE shows competitive performance against other attacks, while its resistance to JPEG compression is slightly weaker. This superior balance of image quality and robustness stems from the *AttentionModule*'s adaptive watermark strength allocation and MoE's expert-specific adapter selection, which jointly minimize image distortion while preserving watermark integrity.

In addition, we report the time for watermarking to reflect the efficiency of the methods. Post-hoc methods tend to take more time, especially optimization-based methods such as SSL. For AWMA-MoE, even with MoE, there is no obvious increase in embedding time, thus preserving the efficiency of in-generation methods.

3.3 Ablation Study

To verify the effect of *AttentionModule* and MoE, we examine four combinations, with or without the two modules, under identical experimental conditions. The results presented in Table 2 show that both modules have positive impacts. The inclusion of

AttentionModule has significantly improved image quality at the cost of a slight degradation in robustness. The MoE module, which selects better adapters from diverse experts for watermarking, can further promote image quality while maintaining bit accuracy.

3.4 Visualization

Figure 2 provides a visual comparison of the watermarked and residual images from in-generation methods. AWMA-MoE achieves the highest PSNR and SSIM for the image, and it has the cleanest residual image. In Figure 3, we show the generated multilevel masks. The similarity between the distribution of masks and the residual image validates the guiding effect of the *AttentionModules*.

4 Conclusion

In this paper, we present AWMA-MoE, a method for embedding flexible watermarks during image generation. AWMA-MoE serves as a plugin for the LDM decoder and utilizes a pre-trained watermark decoder. With the designed attention module and MoE, it can significantly enhance the quality of the generated watermarked images while preserving comparable watermark robustness.

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